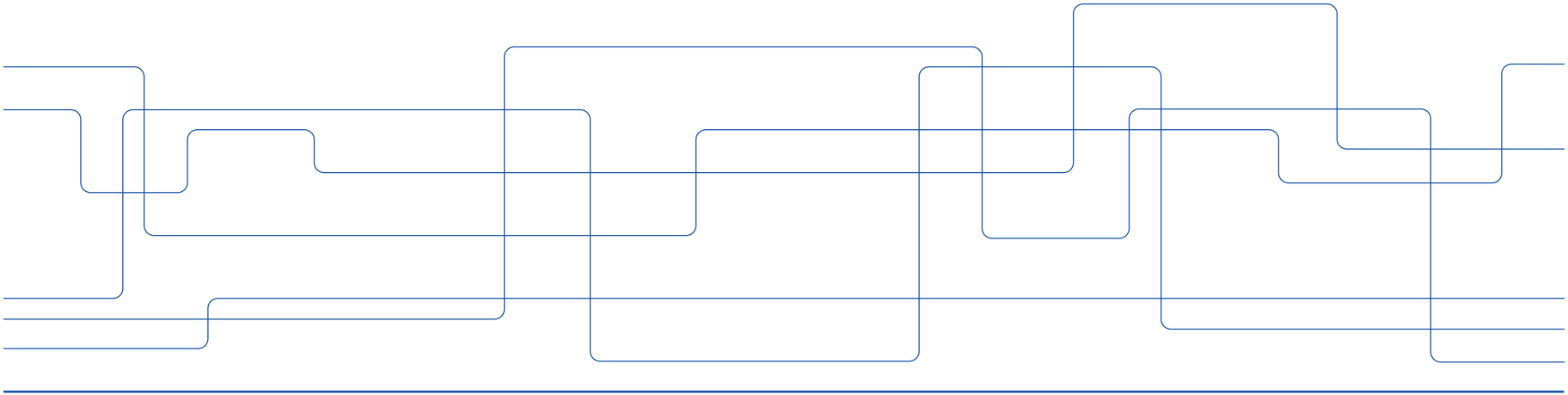
MH2000 Experimental Methods

Thermodynamic Measurements

Dr. Sudhanshu Kuthe



Thermodynamic measurements

- Lecture 1: Isopiestic/Isobaric methods, transpiration method, gas mixtures for controlled chemical potentials, liquid–liquid chemical equilibria

Room: B25

- Lecture 2: Gibbs energy function (gef), second- and third-law methods, electrochemical techniques

Room: D33

Thermodynamic measurements

- ❑ Lecture 1: Isopiestic/Isobaric methods, transpiration method, gas mixtures for controlled chemical potentials, liquid–liquid chemical equilibria

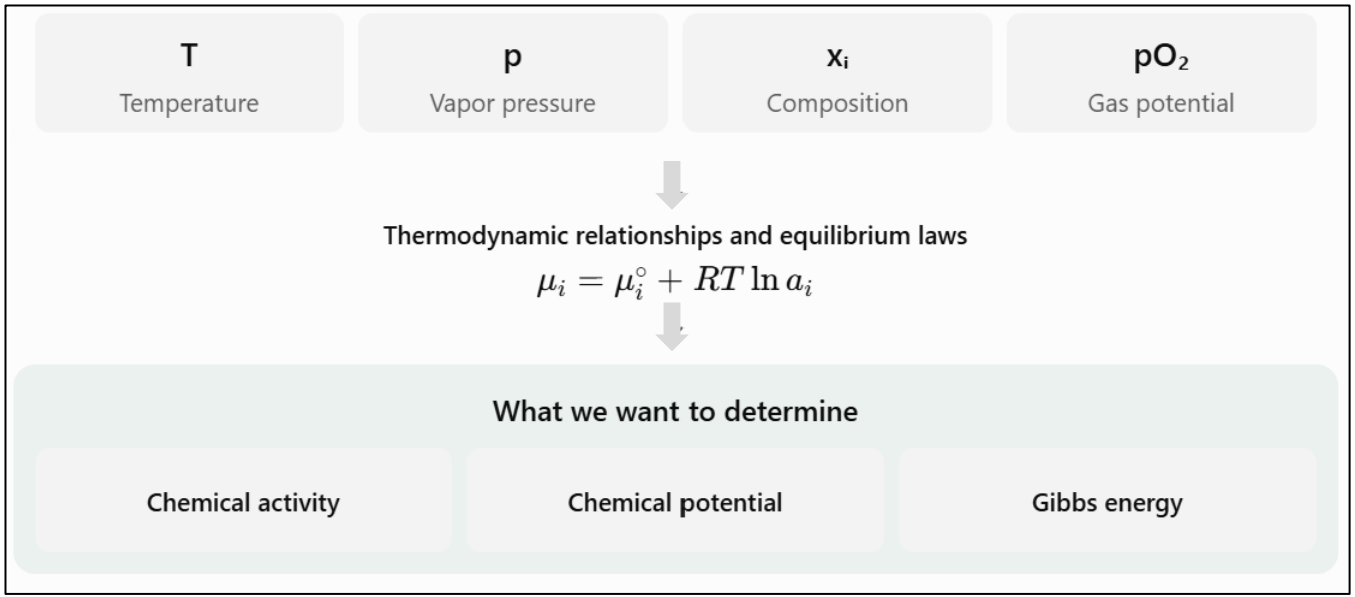
Room: B25

- ❑ Lecture 2: Gibbs energy function (gef), second- and third-law methods, electrochemical techniques

Room: D33

“*How to measure Thermodynamic properties??*

“*What can be measure experimentally??*



“*What can be measure experimentally*??

If we can measure temperature and composition, how can we determine the chemical activity of an element without measuring its Gibbs energy directly.

What is chemical potential?
Chemical potential determines the direction of component transfer. A component tends to move from higher chemical potential to lower chemical potential until equilibrium is established.

What is activity?
Activity describes the effective thermodynamic concentration of a component.

Thermodynamic measurements

- Vapor pressure methods
  - Knudsen effusion
  - Free vaporization methods
  - **Isopiestic/Isobaric methods**
  - **Transpiration method**
  - Mass spectrometry

$$p_i, p_i^\circ, a_i$$

- Calorimetry
  - Reaction calorimetry
  - Bomb calorimetry

$$C_P, \Delta_f H, \Delta_r H, \Delta_r S$$

- Chemical equilibria
  - **Gas-condensed phase equilibria**
  - **Liquid-liquid chemical equilibria**
  - **Electrochemical measurements**

$$a_i, \Delta G, \Delta H, \Delta S, \Delta_{\text{mix}} G, \Delta_{\text{mix}} H, \Delta_{\text{mix}} S$$

- Phase equilibria

Thermodynamic measurements

- Vapor pressure methods
  - Knudsen effusion
  - Free vaporization methods
  - **Isopiestic/Isobaric methods**
  - **Transpiration method**
  - Mass spectrometry

$$p_i, p_i^\circ, a_i$$

- Calorimetry
  - Reaction calorimetry
  - Bomb calorimetry

$$C_P, \Delta_f H, \Delta_r H, \Delta_r S$$

- Chemical equilibria
  - **Gas-condensed phase equilibria**
  - **Liquid-liquid chemical equilibria**
  - **Electrochemical measurements**

$$a_i, \Delta G, \Delta H, \Delta S, \Delta_{\text{mix}} G, \Delta_{\text{mix}} H, \Delta_{\text{mix}} S$$

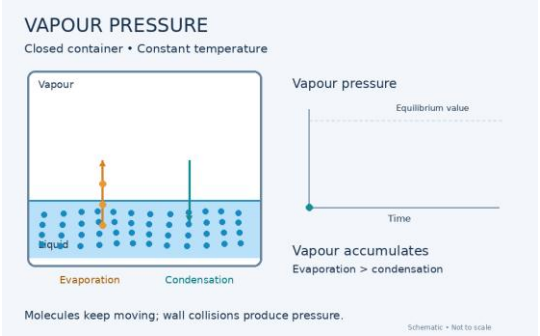
- Phase equilibria

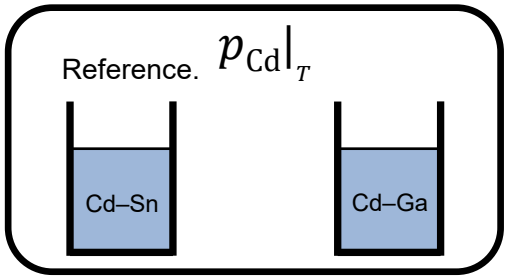
Open vs. closed equilibrium methods

- Closed methods
  - Isopiestic/Isobaric method
- Open circuit methods
  - Transpiration method
  - Gas mixtures for controlled chemical potentials
- Closed circuit methods

Isopiestic/Isobaric methods

- One of the most accurate methods for determining vapor pressures
- Two or more samples are equilibrated at a common vapor pressure of a particular component in a closed vessel
- Has been widely used in aqueous solution chemistry (introduced in 1884)
- Simplest high-temperature system is a sealed capsule under isopiestic/isobaric and isothermal conditions
  - Elements can be transported in the gas phase from one condensed phase to another as metallic vapor, oxide vapor, or chloride vapor
  - Requires a large difference in volatility between solute and solvent

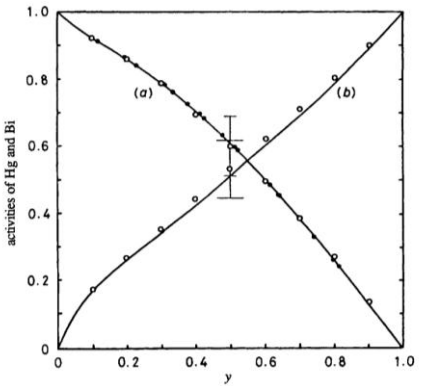


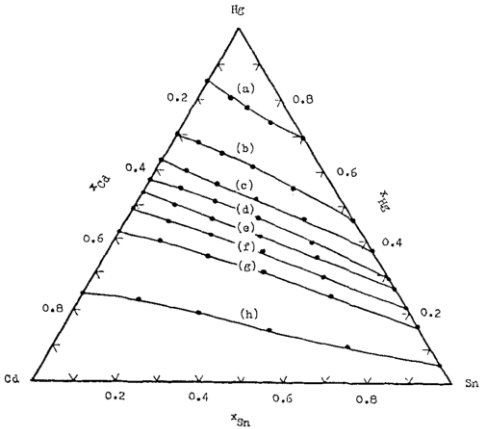


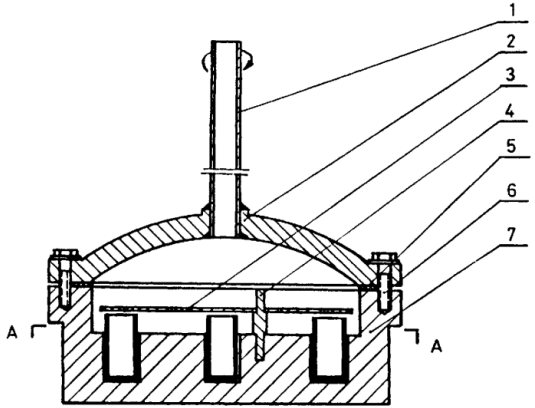
$Temperature$ = constant

Isopiestic/Isobaric methods

- Example: Measured activities at 600 K in Hg–Bi, Hg–Cd, Hg–Ga, Hg–In and Hg–Sn and Hg–Cd–Sn alloys (amalgams)
- Equilibrium attained after 14–24 h







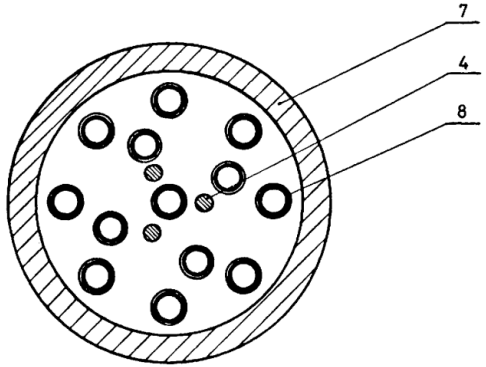
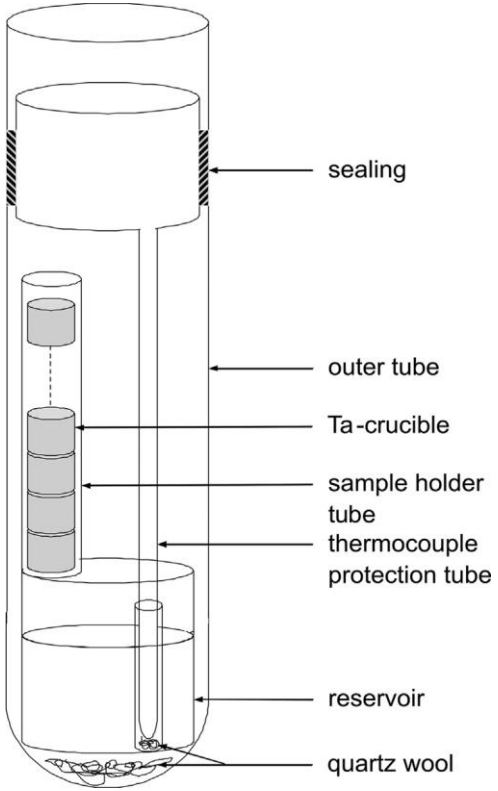


Fig. 1. Experimental arrangement used for measuring the mercury vapour pressure: (1), (2), (7) isopiestic apparatus, (3) stainless-steel cover, (4), (6) screws, (5) sealing washer and (8) graphite crucible.

J. Chem. Soc., Faraday Trans. 1, 1988, 84(12), 4369–4376;
MMTB Vol. 23B, 1992, 623

Pseudoisopiestic method

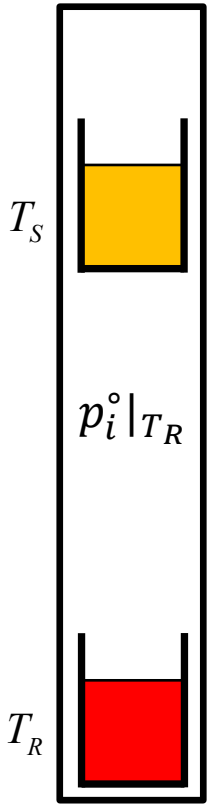
- Sealed vessel with a temperature gradient (non-isothermal)
  - Mostly been used in organic chemistry, but has found limited use in chemical metallurgy
  - Establishment of a fixed vapor pressure above a solution
  - Vapor must not react with the tube or crucible materials
  - Requires reliable data for vapor pressures
  - Have used this method at KTH to study activities in the Mg–Fe, Ca–Fe, and Ca–Si systems



Example: Thermodynamic study of the Ce–Cd system

March 2014
Calphad 44:14–20
DOI: 10.1016/j.calphad.2013.07.005

Pseudoisopiestic method



- Sealed tube, must not let vapor escape
- A pure component in a reservoir exerts an equilibrium vapor pressure that is fixed at a fixed temperature
- An alloy sample is held at a higher temperature in the same closed tube:
  - At equilibrium, the vapor pressure above the alloy will be the same as the vapor pressure in the tube

$$a_i = \frac{p_i^\circ\big|_{T_R}}{p_i^\circ\big|_{T_S}}$$

Pseudoisopiestic method

- Example: The thermodynamic properties of the Au−Cd System
  - The gold absorbs cadmium until the cadmium vapor pressure of the alloy is the same as that of the pure cadmium at the temperature minimum

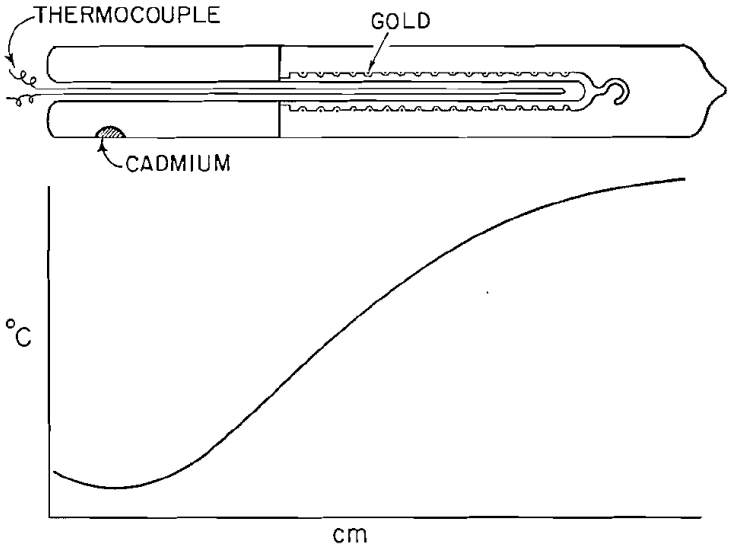
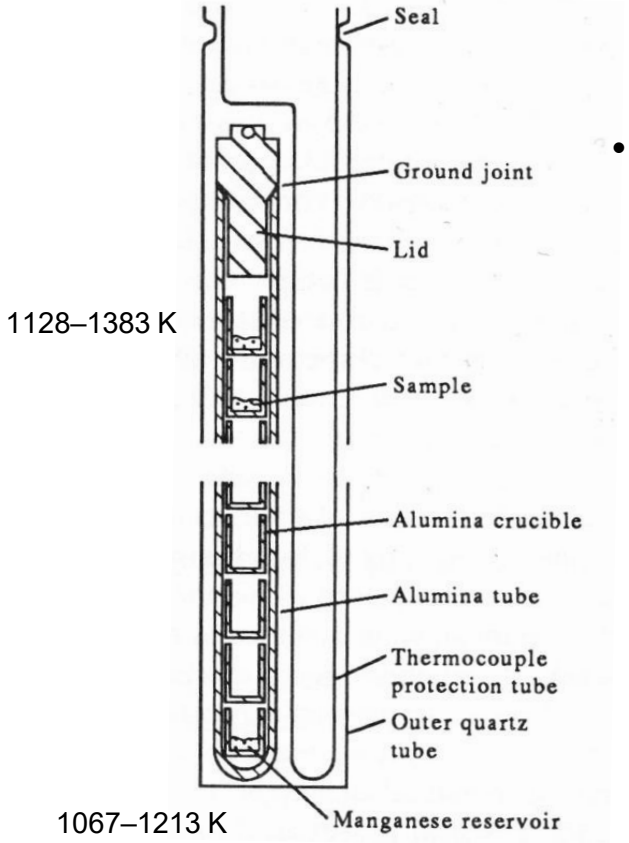


FIG. 1. Quartz vessel and temperature distribution.

Bartha, Alexander: *Canadian Journal of Chemistry*, Vol. 43, 1965

Pseudoisopiestic method

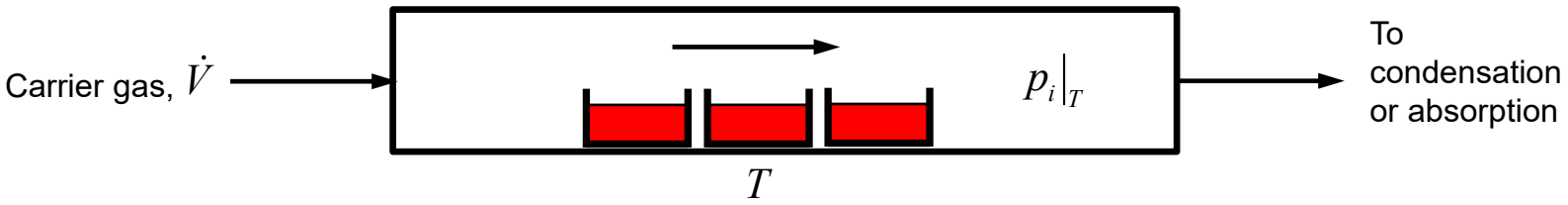


Krachler *et al.*, *Z. Metallkunde*, 1982, **73**,731

- Example: measurement of activity in the Au–Mn system
  - 2-zone resistance furnace
  - Outer tube of quartz evacuated to 10^{-1} Pa and backfilled with pure Ar 3 times, and sealed under vacuum
  - Inner closed tube and crucibles of alumina (Mn reacts with quartz!)
    - > *20 stacked crucibles, bottom one contained pure Mn*
    - > *Ground stopper made to slide into joint by raising assembly into vertical position to seal inner tube*
  - Equilibration time of 6 to 8 weeks!!

Transpiration method

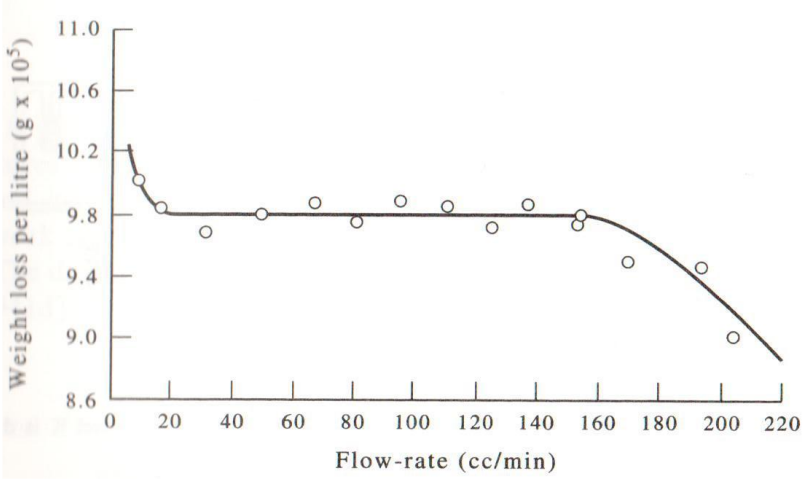
- Entrainment of vapors in an inert or reactive carrier gas
- One of the oldest experimental techniques (1845)
- Excellent accuracy
- Saturation of a steady, measured stream of “carrier gas” with a vapor



- Collection of vapor and calculation of amount
- Rate of vapor removal measured at different gas flow rates

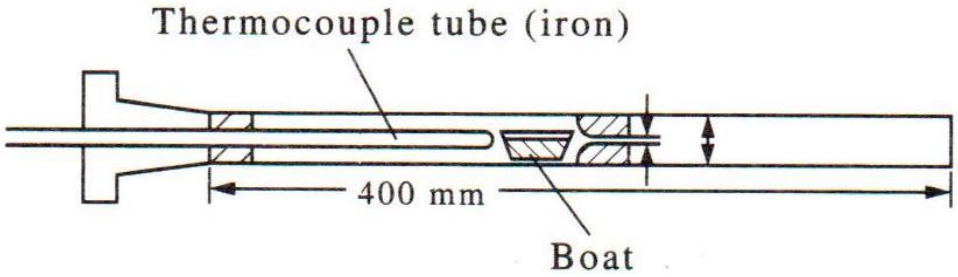
Transpiration method

- In a completely isothermal system, equilibrium partial pressure (saturation) is attained at zero gas flow rate
- In reality, saturation of vapor in a carrier gas can be achieved at low but finite flow rates
  - A plateau can be achieved where the apparent vapor pressure is independent of the carrier gas flow rate
  - At high flow rates, the carrier gas does not have a long enough residence time to become saturated



Transpiration method

- Carrier gas must be quite pure
- For lighter gases, constrictions are made on both sides of the sample to minimize counter-diffusion
  - Argon is a common carrier gas

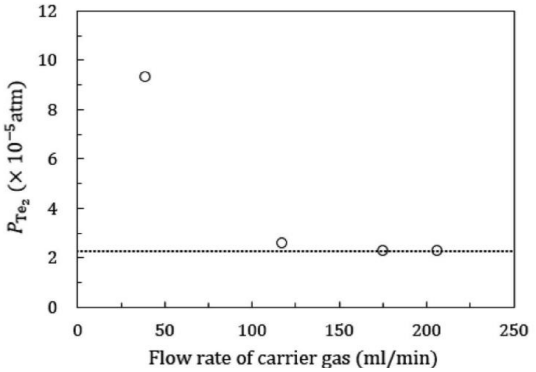


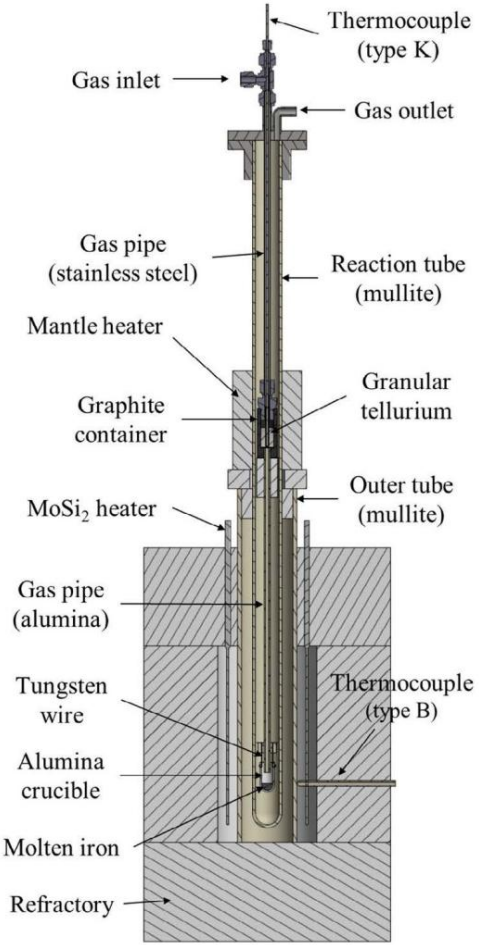
Transpiration method

- Example: Thermodynamic properties of Te in molten Fe
  - 666 K
  - Hydrogen carrier gas
  - Equilibrated 16–32 h

$$\mathrm{Te(g)} = \underline{\mathrm{Te}}(1\ \mathrm{mass\%}\ \mathrm{in\ Fe})$$

$$\Delta G^\circ = (-147\,000 \pm 6\,000) + (53.6 \pm 3.2)\,T\ \mathrm{J/mol}$$





ISIJ International, Vol. 57 (2017), No. 3, pp. 397–403

Transpiration method

- Saturation of the carrier gas can be achieved by bubbling or blowing the carrier gas through a liquid phase
  - Morris and Zellars measured the vapor pressure of copper in this manner (1956) from 1450–1600 °C
    - *Impinged helium and argon on liquid copper surface to create ripples*
    - *“Equilibrium” gas sampled of known volume and drawn through a narrow-bore tube to condense*

Transpiration method

- Another variant is where the carrier gas reacts with a condensed phase to form vapor products
- Example: Vapor pressures in the Hg–O–S system from 300 to 1100 K
  - Measured the Gibbs energy of formation of:

$$HgO,\ HgSO_4,\ Hg_2SO_4,\ HgSO_4 \cdot HgO,\ HgSO_4 \cdot 2HgO$$

  - Study by Fredrickson, Hager, 1996
  - Two-zone furnace with reaction tube made of quartz glass
  - Carrier gas was pre-equilibrated to form $SO_3(g)$

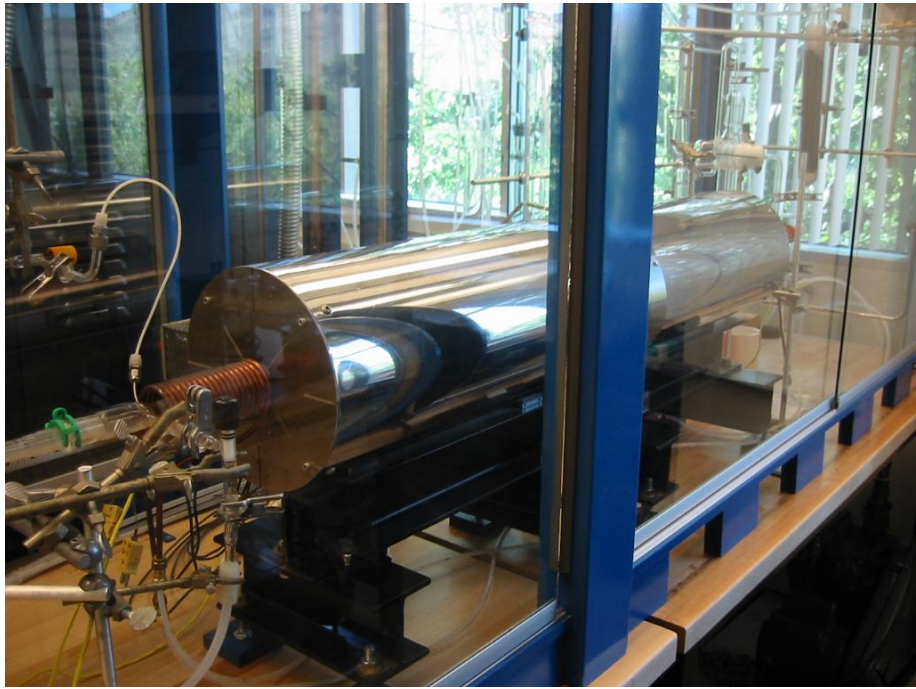
$$SO_2(g) + \frac{1}{2}O_2(g) \xrightarrow[V_2O_5]{} SO_3(g)$$

Transpiration method

- Mercuric sulfate decomposes according to:

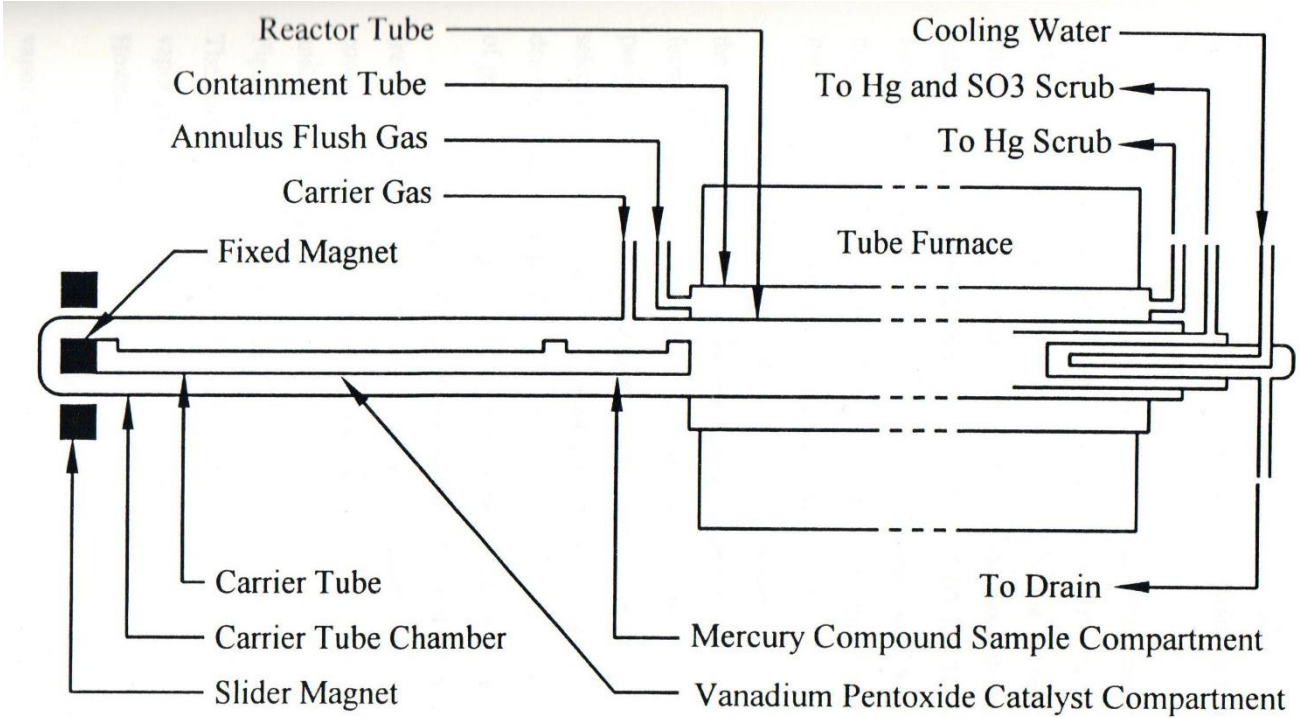
$$HgSO_4 = Hg(g) + SO_3(g) + \frac{1}{2}O_2(g)$$

  - Mercury liquid collected by condensing from the off-gas

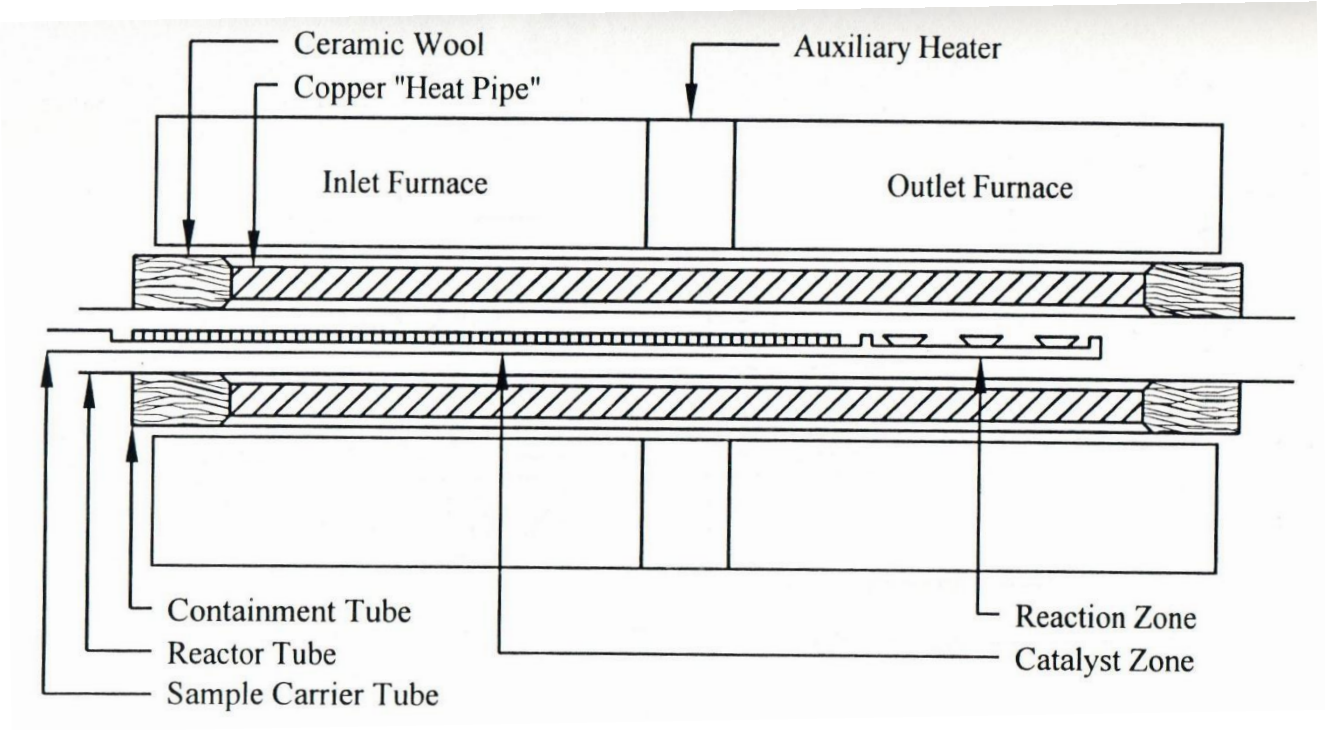


Transpiration apparatus

Transpiration method



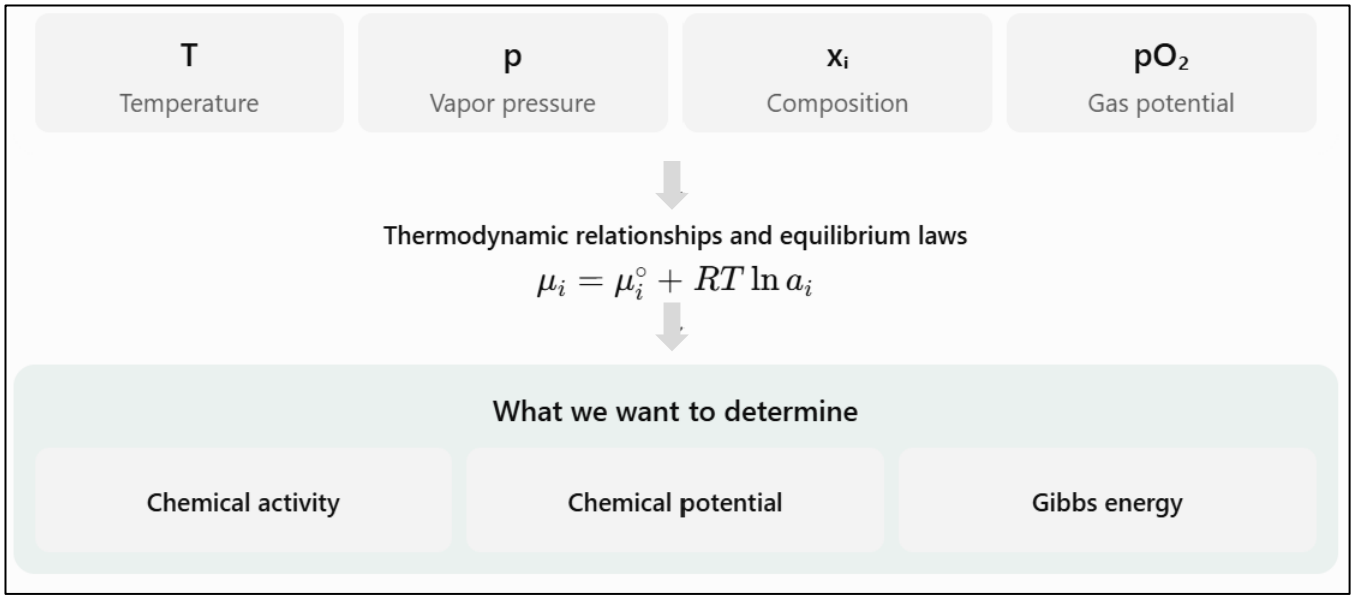
Transpiration method



Gas mixtures for controlled chemical potentials

- Most commonly employed gas mixtures:
  - C–CO–CO_2 equilibria
  - H_2–H_2O equilibria
- Other gas mixtures for controlled chemical potentials:
  - Sulfur potential:
    - > H_2–H_2S gas mixtures
  - Hydrogen potential:
    - > H_2–NH_3 gas mixtures
  - Combined oxygen and sulfur potential:
    - > H_2–H_2S–H_2O
    - > H_2–CO_2–SO_2

“*What can be measure experimentally??*



Why do we need to control oxygen potential?

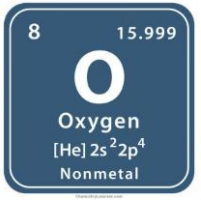
Why do we need to control oxygen potential?

We know that oxygen reacts with liquid metals. But how can we create an extremely low and precisely controlled oxygen partial pressure inside a furnace at 1600 °C

C–CO–CO_2 equilibria







- There are two gaseous oxides of C:

$$C(gr) + O_2(g) = CO_2(g) \qquad (1)$$

$$\Delta G°(1) = -394100 - 0.84T \quad [J]$$

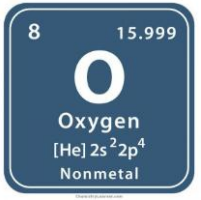
$$2C(gr) + O_2(g) = 2CO(g) \qquad (2)$$

$$\Delta G°(2) = -223400 - 175.3T \quad [J]$$

C–CO–CO_2 equilibria







- A combination of reactions (1) and (2) yields

$$2CO(g) + O_2(g) = 2CO_2(g) \qquad (3)$$

$$\Delta G°(3) = 2\Delta G°(1) - \Delta G°(2) = -564800 + 173.62T \quad [J]$$

- A different combination of reactions (1) and (2) gives

Boudouard Reaction

$$C(gr) + CO_2(g) = 2CO\ (g) \qquad (4)$$

$$\Delta G°(4) = \Delta G°(2) - \Delta G°(1) = 170700 - 174.5T\ [J]$$

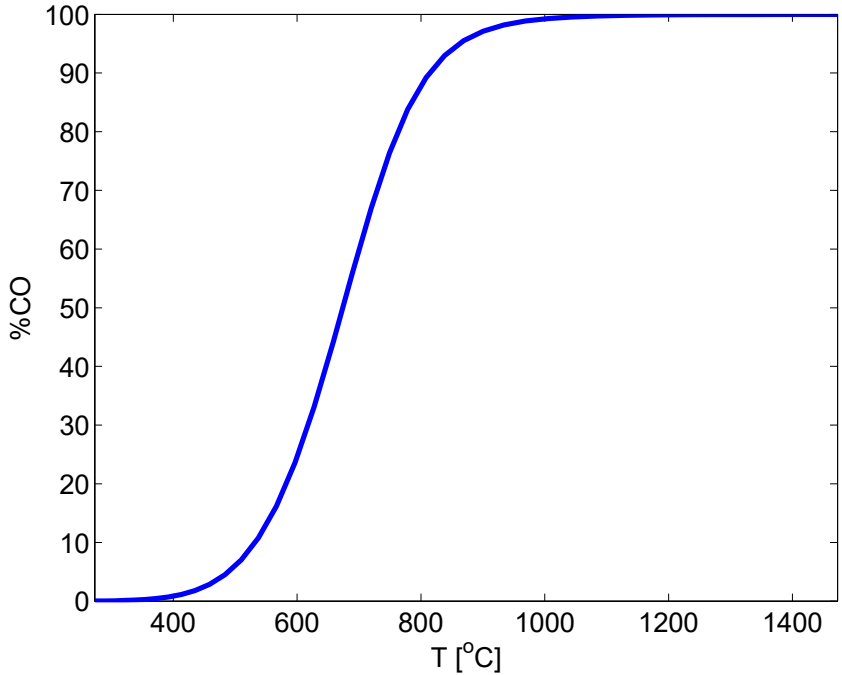
C–CO–CO_2 equilibria



- Can construct a plot %CO vs. *T* with 2 equations

$$p_{CO} + p_{CO_2} = 1$$

$$\Delta G = \Delta G^\circ_{(4)} + RT \ln\left(\frac{p_{CO}{}^2}{p_{CO_2}}\right)$$



Composition of a CO–CO_2 gas mixture in equilibrium with solid graphite, at P_{total} = 1 atm

Controlled oxygen potential with C/CO–CO_2 mixtures

- Below about 1000 °C, a fixed ratio of CO and CO_2 in the inlet gas can be used to control the oxygen potential (in the presence of graphite)
  - Equilibrium partial pressure of CO_2 is significant below 1000 °C

- The oxygen partial pressure is fixed by Rxn 3:

$$2\mathrm{CO(g)} + \mathrm{O_2(g)} = 2\mathrm{CO_2(g)}$$

$$K_3 = \frac{p_{\mathrm{CO_2}}^2}{p_{\mathrm{CO}}^2 \cdot p_{\mathrm{O_2}}} \quad \longrightarrow \quad p_{\mathrm{O_2}} = \left(\frac{p_{\mathrm{CO_2}}}{p_{\mathrm{CO}}} \right)^2 \cdot \frac{1}{K_3}$$

Controlled oxygen potential with C/CO–CO_2 mixtures

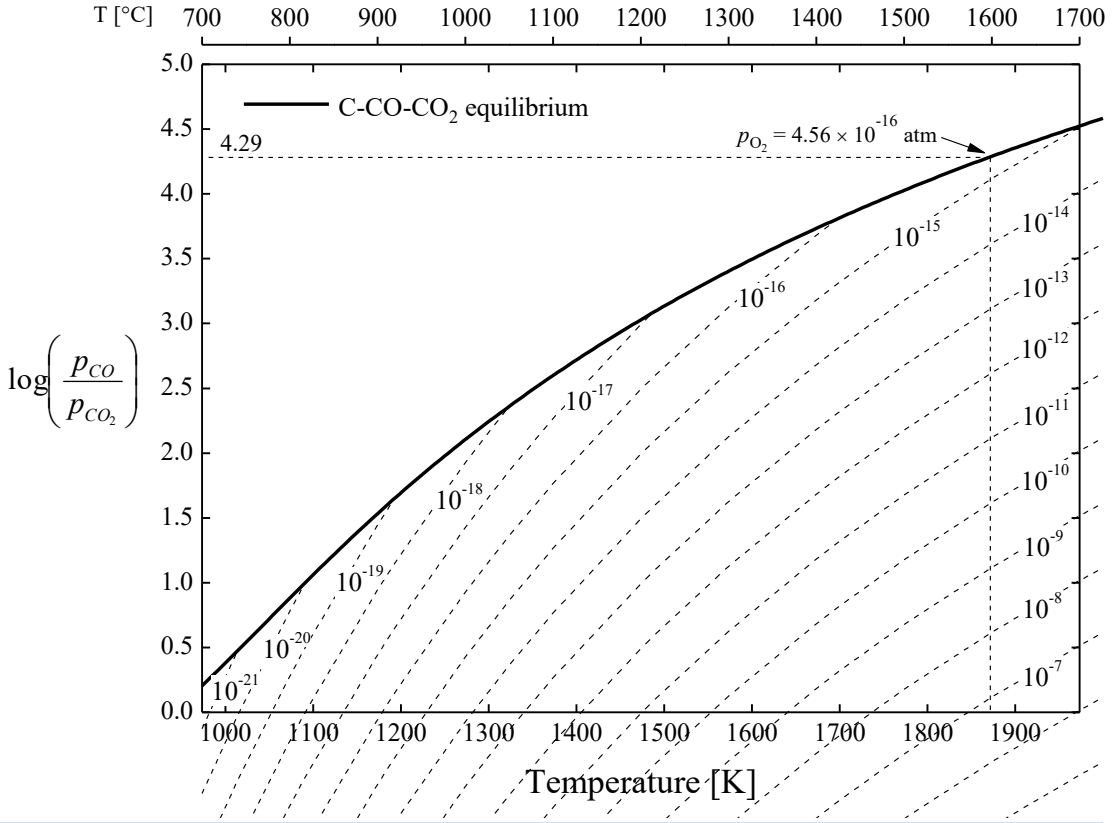
- How can we control the oxygen potential in high temperature experiments above 1000 °C?
  - In equilibrium with graphite - a low range of oxygen partial pressure, ~10^{-16} atm O_2(g)
  - Equilibrium partial pressure of CO_2 is extremely low above 1000 °C
  - Very low p_{O2} – not feasible with oxygen probes
- A fixed oxygen potential can be imposed on the system by fixing the partial pressure of CO in the inlet gas via Rxn 2:

$$2\mathrm{C(gr)} + \mathrm{O_2(g)} = 2\mathrm{CO(g)}$$

$$K_2 = \frac{p_{\mathrm{CO}}^2}{a_{\mathrm{C}}^2 \cdot p_{\mathrm{O_2}}} \longrightarrow p_{\mathrm{O_2}} = \frac{p_{\mathrm{CO}}^2}{K_2}$$

Controlled oxygen potential with C/CO–CO_2 mixtures

- Equilibrium partial pressures of CO and CO_2(g) in equilibrium with graphite with increasing temperature at 1 atm total pressure



Controlled oxygen potential with C/CO–CO_2 mixtures

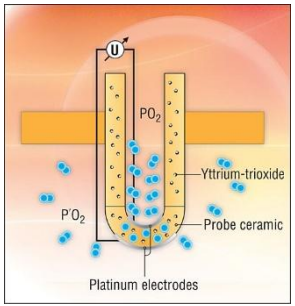
- As can be seen, the equilibrium CO/CO_2 ratio increases with increasing temperature
- For example, at 1873 K, CO is predominant:

$$\log\left(\frac{p_{CO}}{p_{CO_2}}\right) = 4.29 \longrightarrow p_{CO_2} = 5.14 \times 10^{-5}\ \text{atm}$$

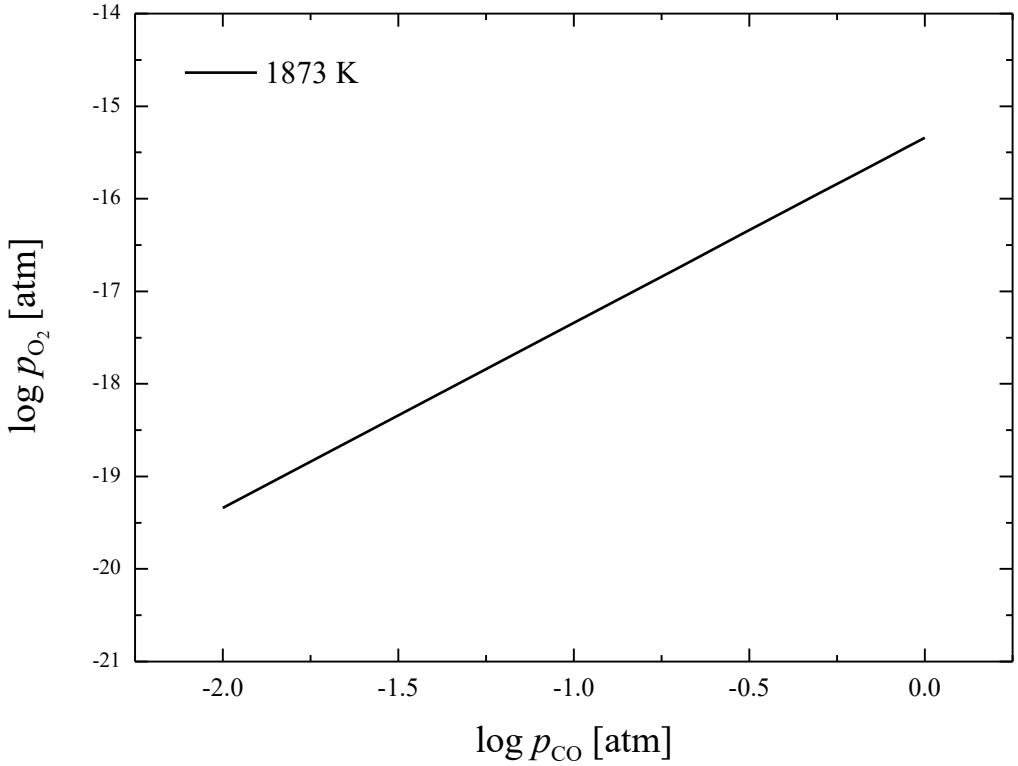
- Any addition of CO_2 will react with graphite to form CO until the CO_2 concentration is lowered to this low equilibrium partial pressure

Controlled oxygen potential with C/CO–CO_2 mixtures

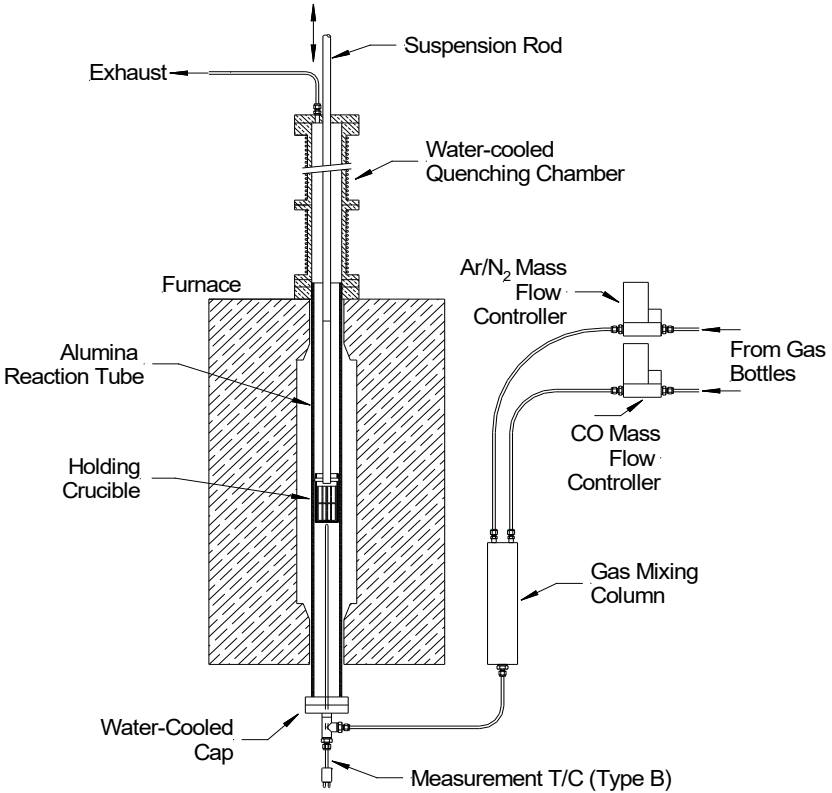
- The partial pressure of O_2 as related to the partial pressure of CO as calculated from Rxn 2



Oxygen ”probe”



Controlled oxygen potential with C/CO–CO_2 mixtures



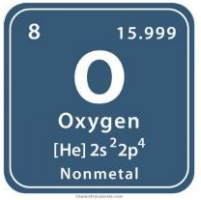


Experimental apparatus at KTH

H_2–H_2O equilibrium







- The equilibrium between hydrogen and oxygen can also be exploited to control the oxygen potential in a system

$$H_2(g) + \frac{1}{2}O_2(g) = H_2O(g)$$

$$K = \frac{p_{H_2O(g)}}{p_{H_2(g)} \cdot p_{O_2(g)}^{1/2}}$$

- H_2/H_2O gas mixtures for controlled oxygen potential
  - An in-line water heater can saturate a hydrogen-bearing carrier gas with water vapor (bubbler)
  - Several experimental designs have been used here at KTH

Controlled oxygen potential with H_2–H_2O mixtures

Gokcen, Chipman, *J. Metals, Trans AIME*, 1952, **194**, 171

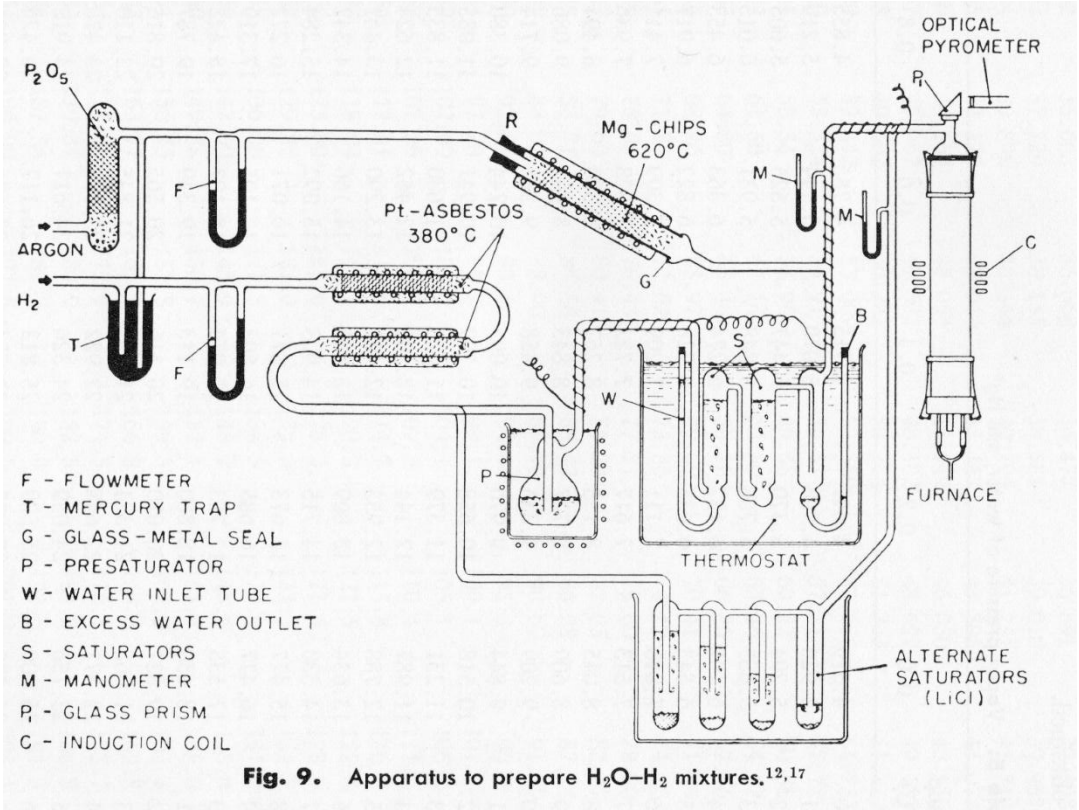
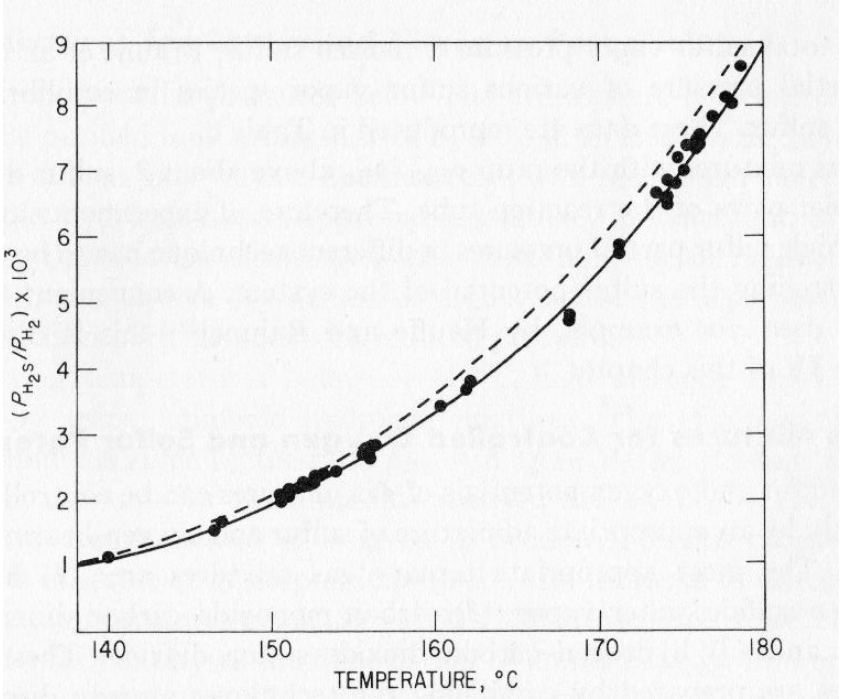


Fig. 9. Apparatus to prepare H_2O–H_2 mixtures.[12,17]

Controlled sulfur potential with H_2–H_2S gas mixtures



$$H_2(g) + \frac{1}{2}S_2(g) = H_2S(g)$$

$$K = \frac{p_{H_2S(g)}}{p_{H_2(g)} \cdot p_{S_2(g)}^{1/2}}$$

Controlled sulfur potential with H_2–H_2S gas mixtures

Rosenkvist, Dunicz, *J. Metals, Trans AIME*, 1952, **194**, 604

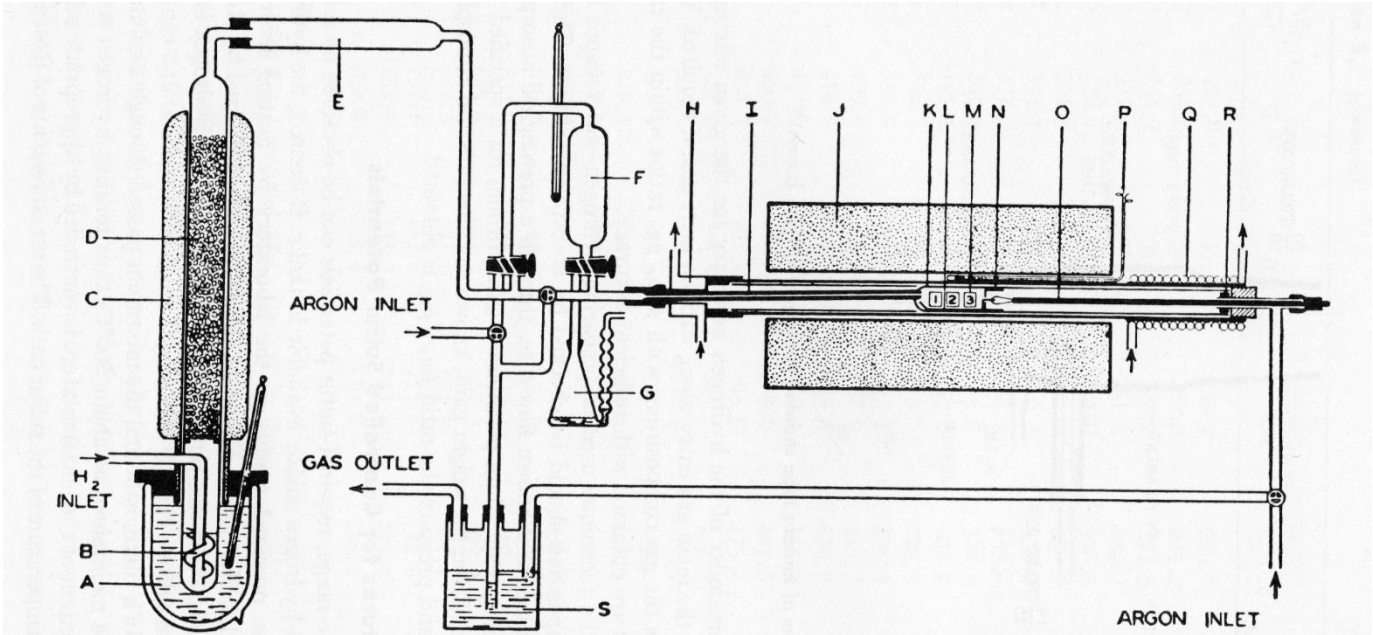
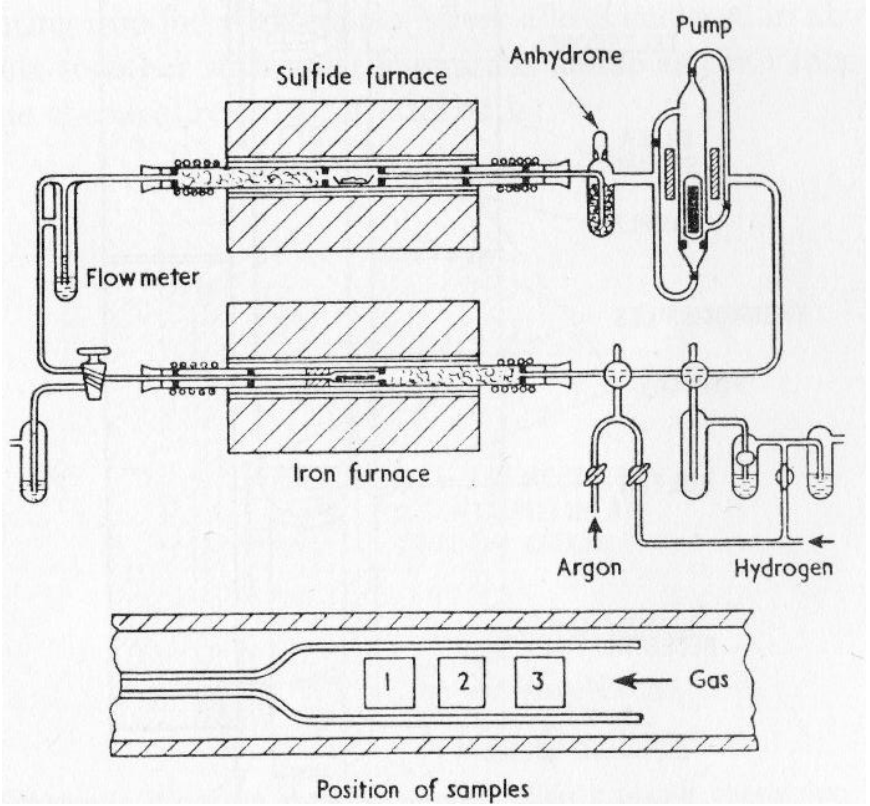


Fig. 11. Apparatus to prepare H_2S–H_2 mixtures.[20] (*A*) Silicone oil thermostat. (*B*) Sulfur bath. (*C*) Furnace, 500°C. (*D*) Alundum chips. (*E*) P_2O_5. (*F*) Gas sampler, 118.5 cc. (*G*) Gas absorption flash. (*H*) Water cooling, (*I*) Al_2O_3 capillary. (*J*) Reaction furnace. (*K*) Al_2O_3 thimble. (*L*) Al_2O_3 boat. (*M*) Iron samples. (*N*) Al_2O_3 baffle. (*O*) Alundum rod. (*P*) Thermocouple. (*Q*) Water cooling. (*R*) Asbestos plug. (*S*) Phosphoric acid.

Closed circuit methods

- Sulfur solubility in iron alloys
  - FeS in sulfide furnace to set S potential

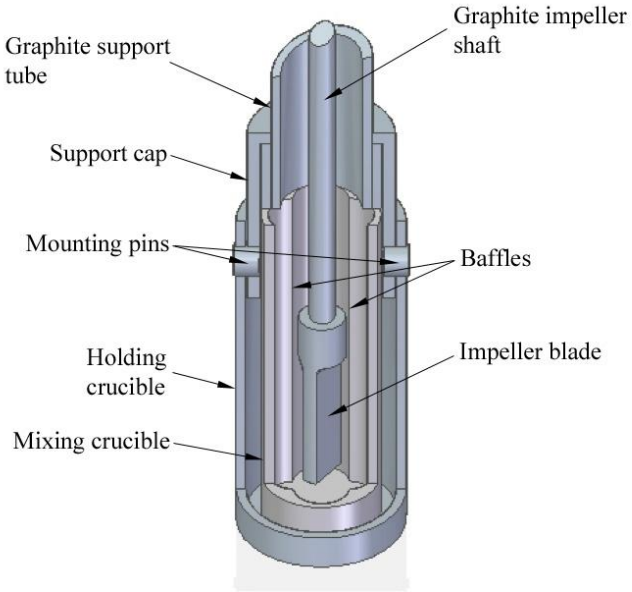
Turkdogan, Ignatowicz, Pearson, *J. Iron Steel Inst.,* 1955, **180**, 349



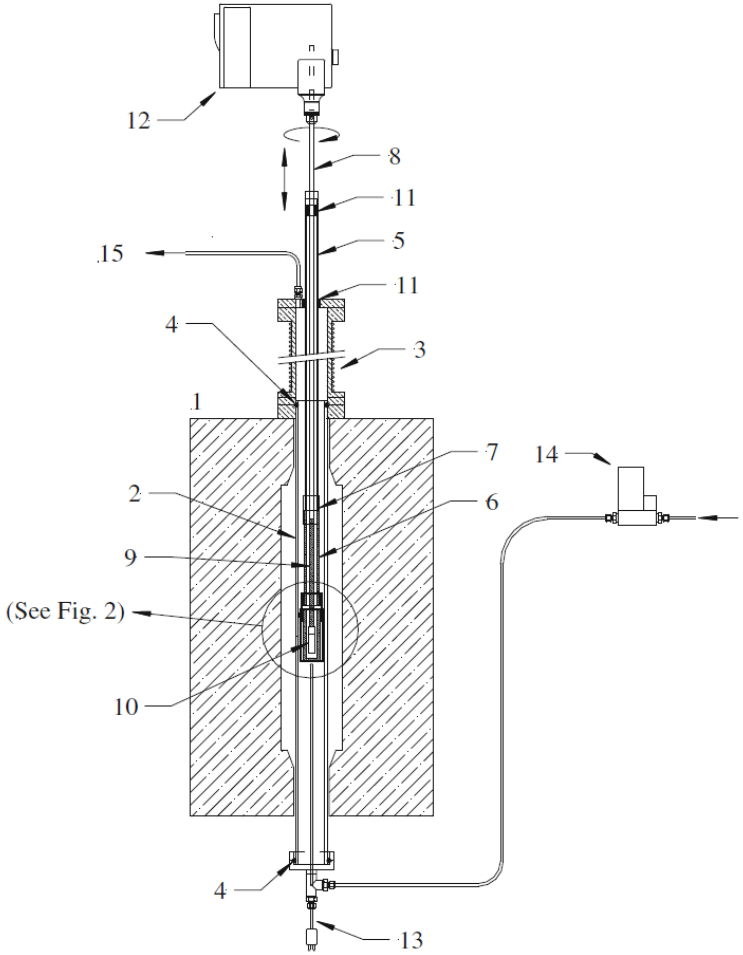
Recirculating system for producing H_2S –H_2 mixtures

Liquid–liquid chemical equilibria

- Induced emulsification



Isometric illustration of the crucible assembly showing parts made of graphite



Impeller mixing experimental setup at KTH

Liquid–liquid chemical equilibria

- Mass transfer of boron from silicon to slag during refining

$$\mathrm{B(l)} + \frac{3}{4}\mathrm{SiO_2(cr)} = \frac{3}{4}\mathrm{Si(l)} + \mathrm{BO_{1.5}(l)} \qquad L_{\mathrm{B}} = \frac{(\%\ \mathrm{B})}{[\%\ \mathrm{B}]}$$

- Mass transfer of calcium during refining

$$\mathrm{CaO(s)} + \frac{1}{2}\mathrm{Si(l)} = \mathrm{Ca(l)} + \frac{1}{2}\mathrm{SiO_2(cr)}$$

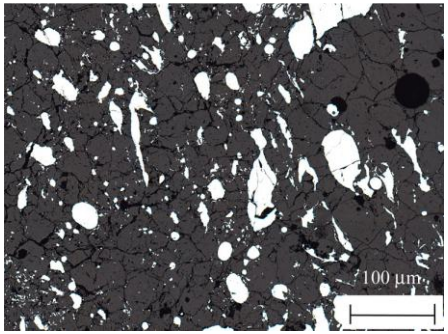
Liquid–liquid chemical equilibria



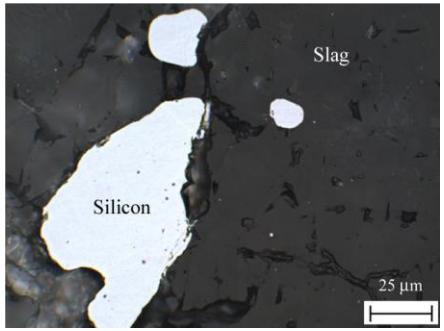
(a) Heat 7, $t = 30$ s (b) Heat 8, $t = 60$ s (c) Heat 24, $t = 180$ s (d) Heat 6, $t = 300$ s

Changes in characteristics of slag–metal interface with increasing reaction time; slag phase doped with boron, 1823 K, 100 rpm mixing speed (White, Sichen 2014)

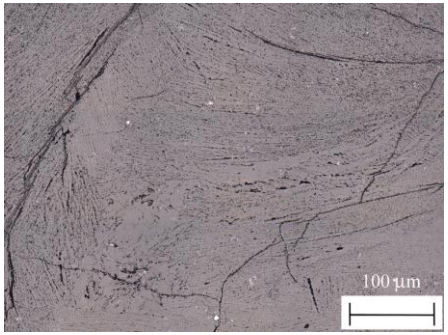
Liquid–liquid chemical equilibria



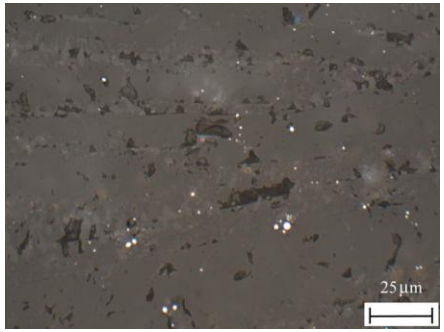
(a) Heat 7, t = 30 s, 50×



(b) Heat 7, t = 30 s, 500×



(c) Heat 11, t = 300 s, 50×



(d) Heat 11, t = 300 s, 500×

Optical micrographs of the slag phase with increasing mixing time showing emulsification (White, Sichen 2014)

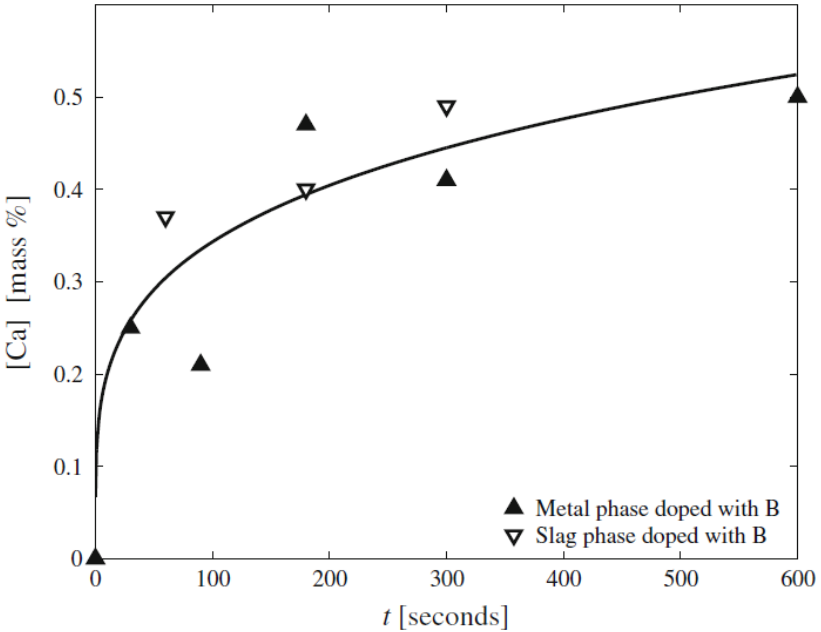
Liquid–liquid chemical equilibria

- Induced emulsification
  - At the initial stage, ΔG of the reaction below is very negative

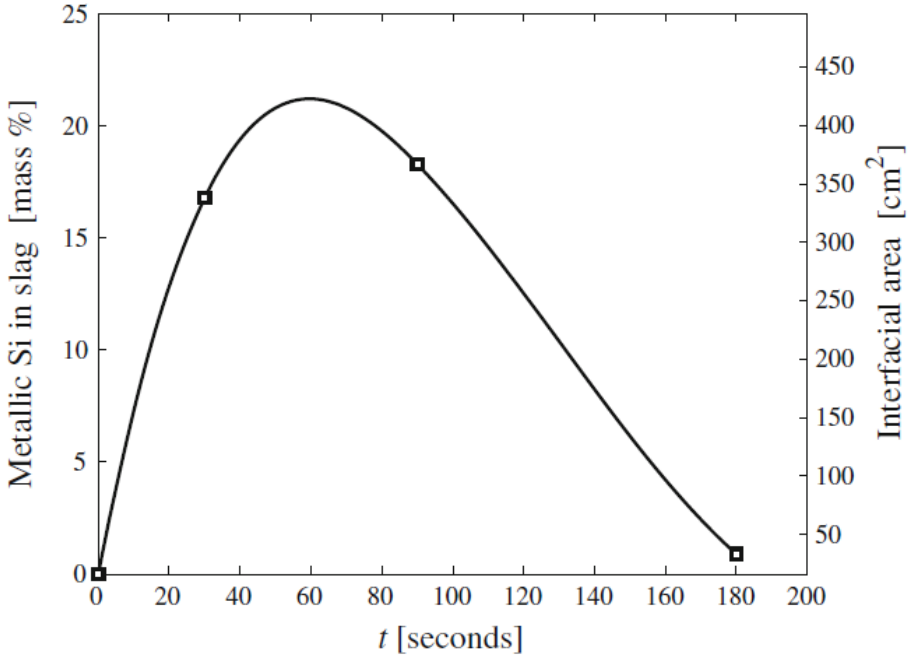
$$[\mathrm{Ca}] + \frac{1}{2}(\mathrm{SiO_2}) = (\mathrm{CaO}) + \frac{1}{2}[\mathrm{Si}]$$

  - The apparent interfacial tension drops due to solute transfer across the interface
  - Mechanical agitation at 100 rpm facilitates the breakup and dispersion of metal phase droplets into the slag phase and greatly increases the interfacial area
  - As the reaction attenuates and ΔG approaches zero, with mixing still going on, there is not enough imparted to the melt to maintain the slag–metal emulsion, and the metal phase recoalesces

Liquid–liquid chemical equilibria



Concentration of calcium in the silicon metal phase with mixing time, 100 rpm, 1823 K (White, Sichen 2014)



Metallic Si present in slag phase and estimation of interfacial area, 100 rpm, 1823 K (White, Sichen 2014)

Summary

Experimental method	What we measure	What we determine
Isopiestic/Isobaric	Equilibrium composition at a common vapor pressure	Chemical activity
Pseudoisopiestic	Temperature and equilibrium alloy composition	Activity of volatile component
Transpiration	Vapor collected and carrier-gas quantity	Vapor pressure and activity
CO-CO_2 / H_2–H_2O	Temperature and gas composition	Oxygen chemical potential
Liquid–liquid equilibrium	Element concentrations in two phases	Distribution ratios and thermodynamic activities, given suitable equilibrium data

References

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2. O. Kubaschewski, C. B. Alcock, P. J. Spencer, Materials Thermochemistry, Sixth Edition, 1993, Permagon Press.

3. G. L. Fredrickson, A Fundamental Thermodynamic Study of Phase Relations in the Mercury-Oxygen Sulfur System at High Temperature, Ph.D. Thesis, Colorado School of Mines, 1996.

4. White, J., Sichen, D.: “Mass Transfer in Slag Refining of Silicon with Mechanical Stirring: Transient Interfacial Phenomena", *Metallurgical and Materials Transactions B*, 2014, vol. 45B, pp. 96–105.